

Online JEPA Composite Stability: A Two-Time-Scale Extension of Borkar (2008) for Recursive Controlled Systems with Learned Substrates

Carlos D. Escobar-Valbuena
Broomva Research
Bogotá, Colombia
carlosdavidescobar@gmail.com

Abstract—Scaffold placeholder. Paper 0 of the Recursive Controlled Systems series proves a per-level stability budget $\lambda_k = \gamma_k - L_\theta \rho_k - L_d \eta_k - \beta \bar{\tau}_k - \frac{\ln \nu}{\tau_a k}$ under hand-set Lyapunov functions and hand-set constants. This paper generalises the budget to the setting in which the substrate V_k itself is *learned online* by a Joint-Embedding Predictive Architecture (JEPA), yielding two new cost terms: an *online encoder cost* $L_{o\theta}(\Delta, \mu)$ derived by extending Borkar’s Ch. 6 two-time-scale stochastic approximation to discrete deployment events, and a *head cost* $L_H \kappa_k$ that bounds substrate sensitivity to head-specific payload variations. The new bound reduces to Paper 0’s Theorem VI exactly in the Paper 0 regime ($\Delta \rightarrow \infty$, $\kappa_k = 0$) — the encoder cost vanishes asymptotically and the head cost is identically zero when no L0 head is engaged at level k . Under an engaged head with $L_H \leq 1$ the head cost is bounded by κ_k but does not vanish; the bound is then strictly tighter than P0 and gates a CI-enforced production rollout via the empirical-constants protocol of Section VI. The final abstract will be written in T5 once the constants table from Q3 substrate measurements is in hand.

Index Terms—stability budget, recursive controlled systems, joint-embedding predictive architecture, two-time-scale stochastic approximation, learned Lyapunov function, deployment-lag analysis, autonomous agents

- [3] C. M. Bishop, *Pattern Recognition and Machine Learning*, ser. Information Science and Statistics. Springer, 2006, ch. 10 (variational inference) is the standard reference for the KL-bound lemma supporting Paper 5 (H8d).

I. INTRODUCTION

II. RELATED WORK

III. RCS RECAP

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V. PROOF

VI. EMPIRICAL-CONSTANTS ESTIMATION PROTOCOL

VII. DISCUSSION

VIII. CONCLUSION

REFERENCES

- [1] V. S. Borkar, *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press / Hindustan Book Agency, 2008, ch. 6 is the foundational two-time-scale stochastic approximation reference; extended by Paper 5 T1 (BRO-989) to discrete deployment events.
- [2] J. N. Tsitsiklis and B. Van Roy, “An analysis of temporal-difference learning with function approximation,” *IEEE Transactions on Automatic Control*, vol. 42, no. 5, pp. 674–690, 1997.